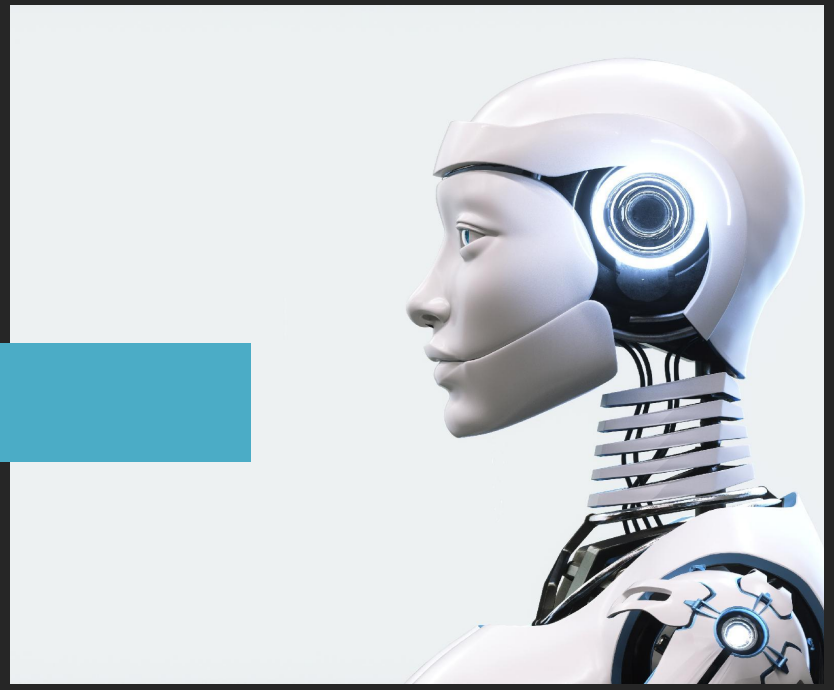
The Impact of AI Permission on Cognitive Performance and Confidence in College Students

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Research Question

- How does explicitly permitting AI use affect college students' CRT-2 logic performance, self-reported AI/outside-assistance use, and their performance confidence?

Background & Support

Past Research Findings:

- High AI reliance can impair critical thinking, often due to **cognitive fatigue** (Tian & Zhang, 2025).
  - However, strong **AI literacy** acts as a buffer against negative cognitive and psychological learning effects (Tian & Zhang, 2025).

What is Cognitive Fatigue?

The fried feeling that happens when you constantly read, tweak, and double-check AI outputs.
Common Signs: brain fog, decision paralysis

AI usage itself isn't detrimental; its effects depend on ***how*** it's used.

We will examine how formal *permission* for AI use alters students' usage, logic capabilities, and self-reported confidence.



Variables



Independent Variables (IVs)

1.) AI Usage Permitted (Between-subjects, manipulated)
-Level 1: AI-Permitted, Level 2: (Control) No AI mention
2.) Gender
-Level 1: Male, Level 2: Female, Level 3: Non-Binary/Prefer not to say
3.) Grouping Variable: Self-Reported Outside Assistance Use

-Level 1: Reported AI Assistance, Level 2: Reported No Assistance



Dependent Variables (DVs)

1.) Actual Performance: Their score on Cognitive Reflection Test (2) (CRT-2) logic puzzles
-4-item logic task scoring actual performance on tricky questions (enforced with timer, minimize AI use)

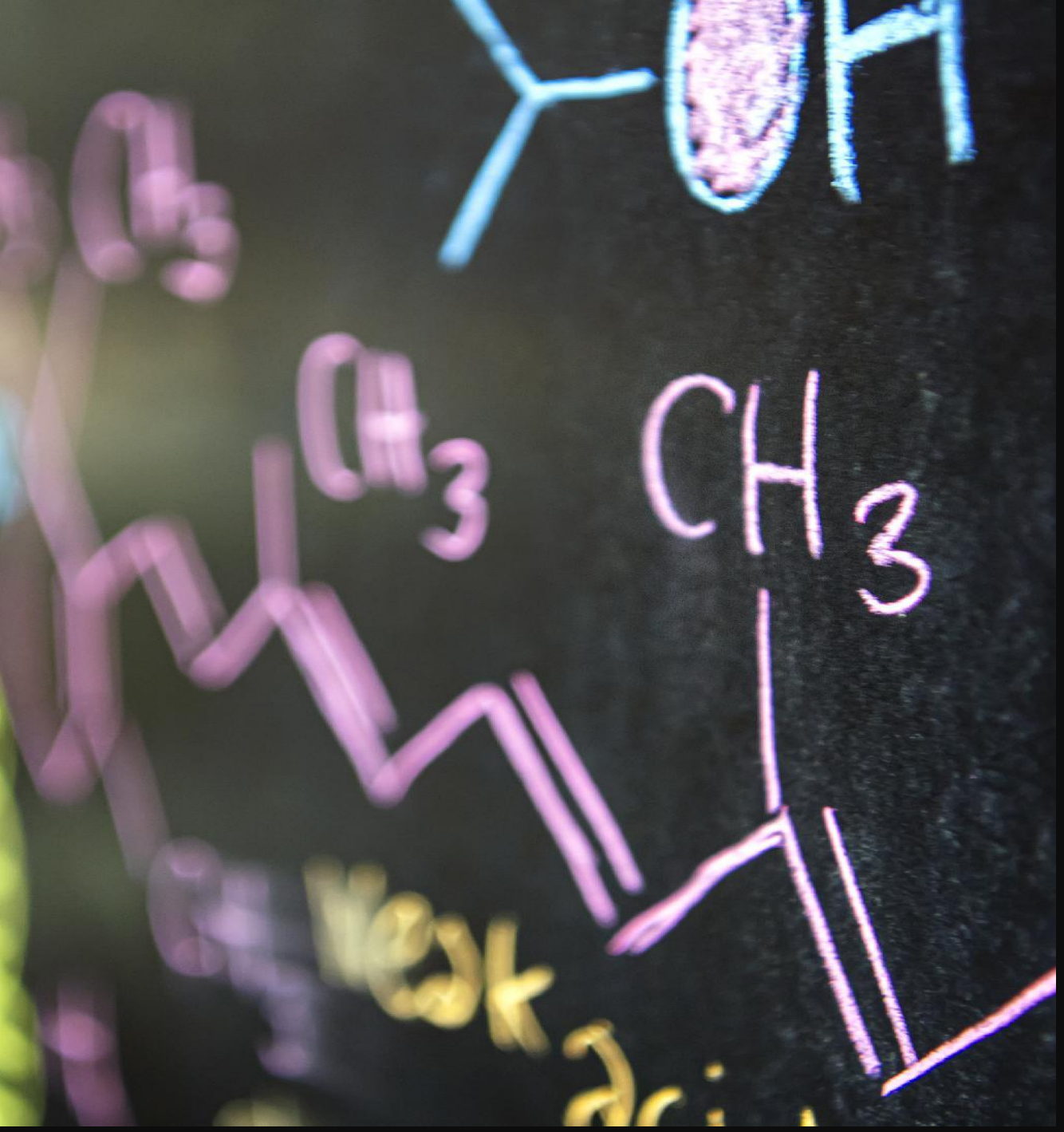
2.) Perceived Performance: Self-rated confidence and performance scale

-Questions asking how confident participant feels they performed



Design

Between-subjects (random assignment to AI Usage condition)



Hypothesis

- Participants in the Group who are explicitly given permission to use AI will be more likely to report using it than those in the control group who were not given any mention of AI use for the logic task.
- Students who report using AI or outside help will score higher on the logic test (CRT) than those who do not (Note: this is a correlational observation, not a causal one.)
- The group permitted to use AI, or who choose to use it, will report higher confidence in their performance.
- We will explore if there are any gender differences in the use of AI, the logic task performance and confidence, to see if there is any significant difference in gender.

References

- Tian, J., & Zhang, R. (2025). Learners' AI dependence and critical thinking: The psychological mechanism of fatigue and the social buffering role of AI literacy. *Acta Psychologica, 260*, Article 105725. https://doi.org/10.1016/j.actpsy.2025.105725